OPERATIONS RESEARCH

Chapter 1 – Multiple Choice Questions

A. Origins of Operations Research

1. What does Operations Research primarily deal with? A. Study of historical operations B. Research on operations within an organization C. Study of only military operations D. Study of financial accounting

2. The emergence of Operations Research was largely associated with: A. Industrial Revolution alone B. World War I C. World War II D. Cold War

3. During World War II, OR was initially developed mainly to: A. Increase employee salaries B. Allocate scarce resources effectively C. Design new weapons only D. Replace military personnel

4. Which countries were specifically mentioned as calling upon scientists for OR work during WWII? A. Germany and Japan B. Britain and the United States C. France and Italy D. Russia and China

5. The first Operations Research teams consisted mainly of: A. Politicians B. Business managers C. Scientists D. Accountants

6. OR teams made an important contribution to the: A. Battle of Britain B. Battle of Waterloo C. Vietnam War D. Korean War

7. One important application of OR during WWII was the development of effective methods for using: A. Sonar only B. Radar C. Satellites D. GPS

B. Growth of Operations Research

10. What was one major reason for the rapid growth of OR after WWII? A. Decline in organizational complexity B. Improvement of OR techniques C. Elimination of computers D. Reduction in mathematical research

11. The simplex method was developed by: A. Isaac Newton B. George Dantzig C. Frederick Taylor D. Henry Ford

12. George Dantzig developed the simplex method in: A. 1937 B. 1942 C. 1947 D. 1957

13. The simplex method is used for solving problems in: A. Linear programming B. Queuing theory only C. Inventory management only D. Probability theory

14. Which of the following were relatively well-developed before the end of the 1950s? A. Linear programming B. Dynamic programming C. Queueing theory D. Inventory theory E. All of the above

15. The computer revolution gave OR a major boost because OR problems often require: A. Very little computation B. Large amounts of computation C. No mathematical calculations D. Only manual calculations

C. Nature of Operations Research

18. OR can be applied to: A. Manufacturing B. Transportation C. Healthcare D. Military operations E. All of the above

19. Which of the following is NOT a characteristic application area of OR mentioned in the chapter? A. Transportation B. Telecommunications C. Financial planning D. Astrology

20. The nature of the organization is, with respect to OR: A. Extremely important because OR applies only to industries B. Essentially immaterial C. Limited to military organizations D. Limited to manufacturing organizations

21. The “research” part of Operations Research implies an approach resembling: A. Guesswork B. Scientific research C. Political decision-making D. Trial without analysis

22. In OR, a hypothesis concerning the real problem is tested through: A. Suitable experiments B. Political voting C. Financial audits only D. Random decisions

23. The process of testing and verifying whether an OR model adequately represents the real problem is called: A. Optimization B. Model validation C. Simulation only D. Resource allocation

D. Organizational Viewpoint

25. OR adopts what kind of viewpoint? A. Individual viewpoint B. Departmental viewpoint only C. Organizational viewpoint D. Political viewpoint

26. The organizational viewpoint attempts to resolve conflicts among components of an organization in a way that is: A. Best for one department B. Best for the organization as a whole C. Best for senior employees D. Cheapest regardless of consequences

27. OR objectives should be consistent with: A. The objectives of a single department B. The objectives of the overall organization C. The objectives of competitors D. Individual employee preferences

E. Optimality

29. OR frequently attempts to find: A. Any possible solution B. The first solution obtained C. An optimal solution D. A random solution

30. Why does the chapter use the phrase “a best solution” rather than “the best solution”? A. OR cannot find good solutions B. There may be multiple solutions tied as best C. OR never uses optimization D. Every problem has only one solution

31. The search for optimality means that OR attempts to: A. Maintain the status quo B. Identify the best possible course of action C. Avoid mathematical models D. Reduce organizational objectives

F. Team Approach

33. Why is a team approach generally necessary for a full-fledged OR study? A. OR problems are always simple B. One person cannot be expected to be an expert in every relevant area C. Computers cannot solve OR problems D. Mathematical knowledge is unnecessary

34. An OR team may require expertise in: A. Mathematics and statistics B. Economics and business administration C. Computer science and engineering D. Behavioral sciences E. All of the above

35. Which of the following is NOT specifically identified as an area of expertise useful to an OR team? A. Probability theory B. Computer science C. Behavioral sciences D. Astrology

G. Impact of Operations Research

37. Operations Research has had a significant impact on: A. Improving organizational efficiency B. Increasing productivity C. Both A and B D. Neither A nor B

38. OR has contributed to increasing the productivity of: A. Only individual companies B. Economies of various countries C. Only military organizations D. Only developing countries

39. IFORS stands for: A. International Federation of Operational Research Societies B. International Forum of Operational Research Studies C. Indian Federation of Operations Research Societies D. International Federation of Operational Resource Systems

H. Applications of OR

41. Which organization used OR to develop national water management policy? A. IBM B. Netherlands Rijkswaterstaat C. United Airlines D. AT&T;

42. Monsanto Corporation used OR to: A. Schedule police officers B. Optimize production operations in chemical plants C. Design aircraft D. Manage hospitals

43. United Airlines used OR to schedule: A. Mining operations B. Shift work at reservation offices and airports C. Oil refinery operations D. Police patrols

44. Citgo Petroleum used OR to optimize: A. Refinery operations and supply/distribution/marketing B. Aircraft schedules C. Hospital beds D. Police patrols

45. The San Francisco Police Department used OR to: A. Design police stations B. Schedule and deploy patrol officers C. Train police officers D. Purchase vehicles

46. Texaco used OR for: A. Optimally blending ingredients into gasoline products B. Scheduling employees C. Managing spare parts D. Designing transportation networks

47. IBM applied OR to: A. Integrate a national network of spare-parts inventories B. Schedule flights C. Manage police patrols D. Optimize gasoline blending

49. Taco Bell used OR to: A. Optimize food recipes B. Schedule employees to provide desired customer service at minimum cost C. Design restaurants D. Manage suppliers only

I. Algorithms and OR Courseware

51. In OR, an algorithm is essentially a: A. Random procedure B. Systematic solution procedure C. Management theory D. Financial statement

52. OR algorithms can efficiently solve problems involving: A. Only a few variables B. Hundreds or thousands of variables C. No variables D. Only two variables

53. The OR Courseware accompanying the book is intended to help students: A. Avoid mathematical models B. Learn and apply OR algorithms C. Replace textbooks D. Study only history

54. OR Tutor is described as: A. A financial software package B. A personal tutor for learning OR algorithms C. A database management system D. A spreadsheet program

55. OR Tutor contains: A. Demonstration examples B. Explanations of algorithms C. Algorithms displayed in action D. All of the above

56. OR Courseware also includes: A. Interactive routines B. Only printed notes C. Only examination papers D. No software

J. OR Software

57. Which of the following is an OR software package mentioned in the chapter? A. LINDO B. LINGO C. CPLEX D. All of the above

58. LINGO is described as a companion to: A. CPLEX B. LINDO C. MPL D. Excel

59. LINDO and LINGO are used in: A. Operations Research B. Image editing C. Video production D. Word processing

60. CPLEX is described as: A. A basic word processor B. An elite state-of-the-art OR software package C. A database D. A presentation program

61. CPLEX is particularly useful for: A. Large and challenging OR problems B. Typing reports C. Drawing diagrams D. Making presentations

62. MPL is primarily described as: A. A user-friendly modeling system B. A spreadsheet C. A database D. A statistical textbook

63. MPL uses which software as its main solver? A. LINGO B. Excel C. CPLEX D. OR Tutor

64. Which software can be used to solve OR examples according to the chapter? A. Excel B. LINDO/LINGO C. MPL/CPLEX D. All of the above

K. High-Importance Exam MCQs

66. OR originated primarily from the need to: A. Increase specialization B. Deal scientifically with complex resource-allocation and operational problems C. Eliminate management D. Replace scientists

67. The beginning of the activity called Operations Research is generally attributed to: A. Industrial Revolution B. World War I C. Military services during World War II D. Computer Revolution

68. Which development was a major milestone in OR? A. Development of simplex method in 1947 B. Development of smartphones in 2007 C. Development of internet in 1990 D. Development of steam engine

69. Which combination correctly represents major characteristics of OR? A. Scientific approach + organizational viewpoint + optimality + team approach B. Guesswork + individual viewpoint + random solutions C. Financial analysis + political viewpoint + trial and error D. Military planning + individual decision-making only

70. Model validation is associated with: A. Testing whether the model/hypothesis is valid for the real problem B. Hiring OR personnel C. Purchasing computers D. Calculating company profit only

71. The primary goal of the organizational viewpoint in OR is to: A. Maximize the benefit of one department B. Resolve conflicts in the best interest of the organization as a whole C. Reduce the number of employees D. Eliminate specialization

72. Which statement about an optimal solution is correct? A. There can be only one optimal solution B. There may be multiple solutions tied as best C. An optimal solution is always the cheapest solution D. OR does not seek optimality

74. Which factor dramatically increased the practical use of OR? A. Decline of computers B. Computer revolution C. Reduction in mathematical techniques D. Elimination of algorithms

75. Which of the following best summarizes Operations Research? A. A method for making decisions using a scientific, analytical and organizational approach to obtain effective/optimal solutions. B. A method used exclusively by the military. C. A branch of accounting concerned only with cost. D. A method of replacing managers with computers.

ANSWER KEY

1. B 2. C 3. B 4. B 5. C 6. A 7. B 10. B 11. B 12. C 13. A 14. E 15. B 18. E 19. D

20. B 21. B 22. A 23. B 25. C 26. B 27. B 29. C 30. B 31. B 33. B 34. E 35. D 37. C 38. B

39. A 41. B 42. B 43. B 44. A 45. B 46. A 47. A 49. B 51. B 52. B 53. B 54. B 55. D 56. A

57. D 58. B 59. A 60. B 61. A 62. A 63. C 64. D 66. B 67. C 68. A 69. A 70. A 71. B 72. B

74. B 75. A